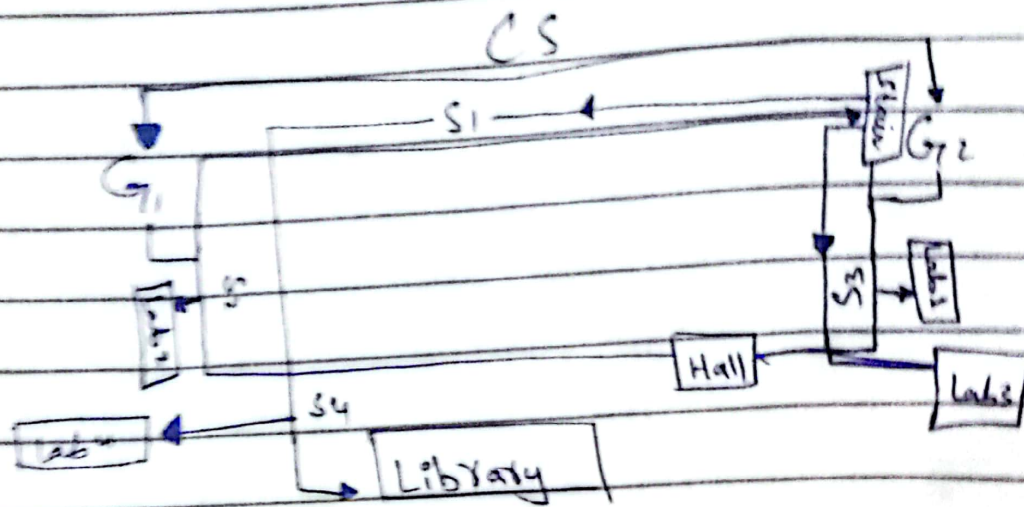


Question: 01



Transition
Table

$CS \rightarrow G_1, G_2$

$G_1 \rightarrow CS, S_1, S_2$

$G_2 \rightarrow CS, \text{stair}, S_3$

$S_1 \rightarrow \text{stair}, G_1$

$S_2 \rightarrow G_1, \text{Lab2}, \text{Hall}$

$\text{stair} \rightarrow G_2, S_1, \text{Lab3}, S_4$

$S_3 \rightarrow G_2, \text{Lab1}, \text{Hall}$

$\text{Hall} \rightarrow \text{Lab3}, S_4$

$S_4 \rightarrow \text{Lab4}, \text{Library}$

Signature _____

RC

No. _____

Question 2

Initial State : Library

Goal state : Hall

State space : CS Department

Action : Running

Transition table : Library to Hall.

Path cost : cost = 1

Question : 04

Shortest path

Library

S₄

Stair

G₁₂

S₃

Hall

Question 3: Find Path Library to Hall?

$$Q = \{ \text{Library} \}$$

$$V = \{ \text{Library} \}$$

Dequeue Library.

$$Q = \{ S_4 \}$$

$$V = \{ \text{Library}, S_4 \}$$

Dequeue S_4

$$Q = \{ \text{Lab4}, \text{stair} \}$$

$$V = \{ \text{Library}, S_4, \text{Lab4}, \text{stair} \}$$

Dequeue Lab4

$$Q = \{ \text{stair} \}$$

$$V = \{ \text{Lib}, S_4, \text{Lab4}, \text{stair} \}$$

Dequeue stair

$$Q = \{ S_1, G_2, \text{Lab3} \}$$

$$V = \{ \text{Lib}, S_4, \text{Lab4}, \text{stair}, S_1, G_2, \text{Lab3} \}$$

Dequeue S_1

$$Q = \{ G_2, \text{Lab3}, G_1 \}$$

$$V = \{ \text{Lib}, S_4, \text{Lab4}, \text{stair}, S_1, G_2, \text{Lab3}, G_1 \}$$

Dequeue Lab G_2

$$Q = \{ \text{Lab3}, G_1, S_3 \}$$

$$V = \{ \text{Lib}, S_4, \text{Lab4}, \text{stair}, S_1, G_2, \text{Lab3}, G_1, S_3 \}$$

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RC

No. _____

Date _____

Dequeue G_1

$Q_2 = \{S_3, S_2\}$

$V_2 = \{Lib, S_4, Lab_4, Stair, S_1, G_2, Lab_3, G_1, S_3, S_2\}$

Dequeue S_3

$Q_2 = \{S_2, Hall\}$

$V_2 = \{Lib, S_4, Lab_4, Stair, S_1, G_2, Lab_3, G_1, S_3, S_2, Hall\}$